Exact Parameter Recovery for Multi-Prism Risley Systems from Beam Scan Patterns via Global Optimisation

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Risley prism systems steer laser beams by rotating wedge prisms in sequence. We address the *parameter-recovery* inverse problem—recovering rotation speeds, wedge angles, glass indices, and system geometry from an observed scan pattern—for systems with $P = 2$–4 prisms (4–8 refractive interfaces, up to 12 free parameters). Unlike beam-steering inverses that assume known hardware, our solver treats the hardware itself as unknown.

A two-stage pipeline combines (i) a multi-task neural network for prism-count classification (98.8 %) with (ii) multi-restart differential evolution, Nelder–Mead polishing, and permutation search. We identify a *speed-based identifiability condition*: prisms sharing the same rotation speed create a degenerate valley in the MSE landscape.

On a 23-case test battery ($P = 2$–4) including adversarial configurations (near-identical speeds, tiny angles, harmonic ratios), all parameters are recovered to floating-point precision (MSE $\sim 10^{-23}$); real-world accuracy is bounded by the fidelity of the forward model to the physical system. A 30-trial-per-SNR noise study shows graceful degradation (100 % success at 60 dB, 57 % at 35 dB). Model-mismatch tests quantify tolerance to geometry errors (< 3 % safe). An NN ablation reveals that the network's value is classification, not regression warm-starting. Out-of-distribution tests show the residual MSE cleanly separates Risley-compatible patterns from incompatible ones. This is a simulation study; experimental validation is planned as future work.

I. INTRODUCTION

A. Background

Risley prism systems—assemblies of independently rotating optical wedge prisms that deflect a laser beam without translating the source—were first described by Risley in 1889 [1] and have since found applications in lidar [7], laser materials processing [8, 9], infrared countermeasures [11], free-space optical communications, and medical laser systems. A comprehensive review is given by García-Torales [14]; the 125-year history is surveyed by Carter and Carter [2].

A physical wedge prism is a solid piece of glass with two non-parallel planar faces. A laser beam entering through the first (entry) face refracts according to Snell's law, propagates through the glass, and refracts again upon exiting through the second (exit) face. Each prism thus contributes *two* refractive interfaces. When the prism rotates, both faces rotate in lockstep as a rigid body. A system of P prisms therefore has $2P$ refractive interfaces.

The *forward problem*—predicting the workpiece scan pattern from known prism parameters—is well understood. Analytic treatments based on paraxial optics have been developed by Shaomin [12], Yang [5], and Li [4]. Numerical ray-tracing valid beyond the paraxial regime has been demonstrated by Marshall [3], Schitea *et al.* [13], and Duma and Dimb [6].

B. The inverse problem

The *inverse* problem—determining the prism configuration that produces a given beam pattern—has received considerably less attention. Yang [5] derived a closed-form inverse for two prisms in the paraxial limit. Ostaszewski *et al.* [10] described look-up-table inversion for two prisms. Bos *et al.* [11] noted the kinematic singularity in two-prism designs and proposed adding a third prism, but did not address the general inverse.

Recent neural-network approaches have targeted the *beam-steering* inverse: given a desired target point, find the prism rotation angles that steer the beam there. Li *et al.* [19] trained a back-propagation neural network (BPNN) to map target coordinates to two-prism rotation angles, achieving arcsec-level accuracy in 80 μs. Yan *et al.* [20] introduced physics-constrained mixture density networks that handle the non-uniqueness of the steering inverse under non-paraxial, thick-prism conditions. These methods assume *known hardware*—the prism geometry, glass type, and wedge angles are given—and solve for the instantaneous rotation angles needed to reach a target.

Our problem is fundamentally different: we assume the beam pattern (a time-series of workpiece positions) is *observed*, and the physical prism parameters themselves—rotation speeds, wedge angles, glass indices, and system geometry—are *unknown*. This is a system-identification problem rather than a beam-steering problem.

For this parameter-recovery inverse with $P \geq 2$ prisms (≥ 4 interfaces), no prior solution has been reported. The difficulty is threefold:

1. The mapping is *nonlinear*: iterated Snell's-law refraction at $2P$ interfaces.
2. The parameter space grows as $3P$ (speed, wedge angle$_x$, wedge angle$_y$ per prism), and the total interface count as $2P$.
3. The inverse may be *degenerate*: different parameter vectors can produce identical patterns when per-prism contributions are not independently observable.

C. Contributions

1. A **two-interface-per-prism physical model** with lockstep rotation and alternating air–glass–air refraction.
2. A **speed-based identifiability condition**, demonstrated via MSE landscape analysis.
3. **Exact recovery (to floating-point limits)** on 23 test cases ($P = 2$–4) including adversarial configurations, with validated Hessian error bounds ($\kappa \approx 2.4 \times 10^7$).
4. **Quantified noise robustness** (30 trials/SNR, reliable to 40 dB) and **model-mismatch tolerance** (<3 % geometry error).
5. **Comparison with prior methods**: the paraxial FFT inverse and beam-steering neural networks [19, 20].
6. An **NN ablation** showing the network's value is classification (98.8 %), not regression warm-starting.
7. A **residual-MSE diagnostic**: the forward map produces quasi-periodic (C^∞) curves; the solver's residual MSE separates Risley-compatible patterns ($<10^{-3}$) from incompatible ones (>1).

II. PHYSICAL PRISM MODEL

A. System geometry

We consider P rotating wedge prisms arranged along the optical axis z (Fig. 1). Each prism i ($i = 1, \ldots, P$) is a solid glass body with refractive index $n_{g,i}$ and two planar faces:

- **Entry face** (interface $2i-1$): flat, perpendicular to z when $\gamma_i = 0$. The beam refracts from air ($n = 1.0$) into glass ($n = n_{g,i}$).

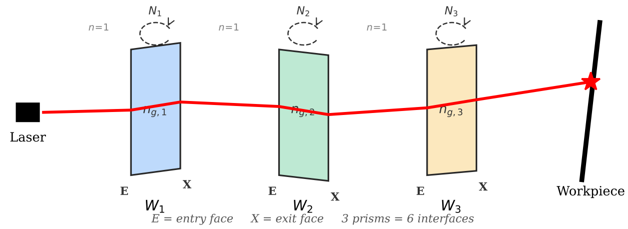

FIG. 1. Three-prism Risley system. Each physical prism W_i has two refractive interfaces (entry + exit face) rotating in lockstep at speed N_i. The beam refracts at each air–glass and glass–air transition.

- **Exit face** (interface $2i$): tilted at the wedge angle $(\alpha_{x,i}, \alpha_{y,i})$. The beam refracts from glass back into air.

Both faces rotate rigidly at speed N_i (Hz), so the rotational angle is

$$\gamma_i(t) = (360\, N_i\, t) \bmod 360^\circ. \tag{1}$$

The refractive indices along the optical path alternate:

$$n_0 = 1.0,\ n_1 = n_{g,1},\ n_2 = 1.0,\ n_3 = n_{g,2},\ n_4 = 1.0,\ \ldots \tag{2}$$

ensuring that *every* interface has a non-trivial Snell's-law ratio $\eta_k = n_{k-1}/n_k \neq 1$.

The beam originates at $(r_x, r_y, 0)$ with initial angles $(\theta_x^{(0)}, \theta_y^{(0)})$. Inter-element distances are: source to first entry face d_0; prism thickness τ; air gap between prisms δ; last exit face to workpiece d_W.

B. Refraction at each interface

At interface k with effective tilt φ_k, the beam direction is updated by vector Snell's law. With unit surface normal $\hat{\boldsymbol{N}} = [\tan\varphi,\, 0,\, -1]^\top/\|\cdot\|$, unit incident direction $\hat{\boldsymbol{s}}_\mathrm{i} = [\tan\theta,\, 0,\, 1]^\top/\|\cdot\|$, and ratio $\eta = n_{k-1}/n_k$:

$$\boldsymbol{s}_\mathrm{f} = \eta\,(\hat{\boldsymbol{N}} \times (-\hat{\boldsymbol{N}} \times \hat{\boldsymbol{s}}_\mathrm{i})) - \hat{\boldsymbol{N}}\,\sqrt{1 - \eta^2\,|\hat{\boldsymbol{N}} \times \hat{\boldsymbol{s}}_\mathrm{i}|^2}. \tag{3}$$

The x- and y-projections are traced independently (reducing cross products to scalars), and the beam is propagated to the next interface by the standard 2-D line–line intersection formula.

C. Effective tilt from rotation

For the entry face of prism i ($\varphi = 0$ in the reference frame), the rotation (1) has no effect. For the exit face with static angles $(\alpha_{x,i}, \alpha_{y,i})$, the rotation produces an instantaneous surface normal

$$\boldsymbol{n}_i = \begin{bmatrix} \cos(\gamma_i + \alpha_{y,i})\, \tan\alpha_{x,i} \\ \sin(\gamma_i + \alpha_{y,i})\, \tan\alpha_{x,i} \\ -1 \end{bmatrix}, \tag{4}$$

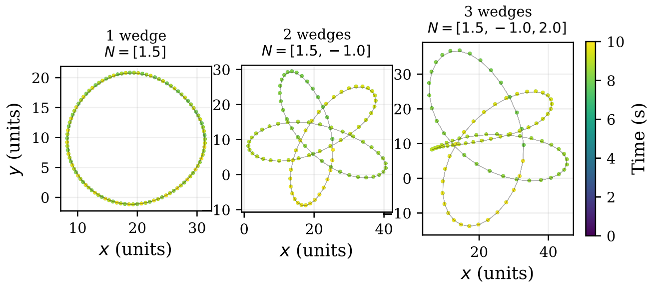

FIG. 2. Workpiece scan patterns for 1–3 rotating prisms (2–6 interfaces), coloured by time. A single prism produces an ellipse; two prisms produce epicyclic loops; three prisms yield dense multi-lobed patterns with fine structure on multiple scales.

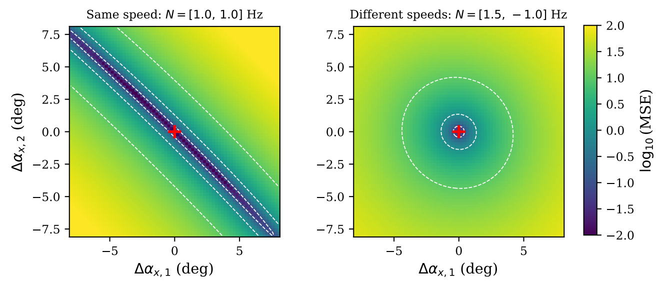

FIG. 3. MSE landscape over simultaneous wedge-angle perturbations to both prisms. Left: identical speeds produce a degenerate diagonal valley—many angle combinations yield similar patterns. Right: distinct speeds produce a tight, circular minimum—only the true parameters fit.

from which effective per-axis tilts are extracted via $\varphi_x = 90^\circ - \arccos(n_x/\|\boldsymbol{n}\|)$ (and similarly for y).

D. Vectorised implementation

Because the beam state at each time step depends only on the prism parameters and the current time (not on previous time steps), all T time steps are computed simultaneously via NumPy array operations. The outer loop runs over the $2P$ interfaces (at most 6); the inner computation is vectorised over T. This yields a $\sim 20{,}000\times$ speedup over a serial reference implementation, generating 200-point patterns in ≈ 0.5 ms ($P = 3$). The vectorised and serial codes agree to $\sim 4 \times 10^{-13}$, confirming numerical consistency; validation against an external ray-tracing package (e.g., Zemax) or physical measurements is deferred to future work.

E. Example patterns

Figure 2 shows scan patterns for one, two, and three prisms. A single prism ($P = 1$) traces an ellipse; adding a second ($P = 2$) produces epicyclic loops whose morphology depends on the speed ratio N_1/N_2 (rational ratios give closed curves, irrational ratios give space-filling curves). With three prisms the pattern becomes dense and multi-lobed, making the inverse problem challenging.

III. IDENTIFIABILITY ANALYSIS

A. The identical-speed degeneracy

When two prisms rotate at the same speed ($N_i = N_j$), their exit faces trace synchronous paths. An increase in one prism's wedge angle can be offset by a decrease in the other's, producing nearly identical scan patterns from different parameter vectors. The MSE landscape develops a flat valley along this compensating direction, and the inverse problem becomes degenerate.

Figure 3 demonstrates this with a two-dimensional MSE landscape. Both panels show $\log_{10}(\text{MSE})$ as a function of simultaneous perturbations $\Delta\alpha_{x,1}$ and $\Delta\alpha_{x,2}$ to the two prisms' wedge angles. The red cross marks the true solution. In the left panel ($N_1 = N_2 = 1.0\,\text{Hz}$), a pronounced diagonal valley runs from upper-left to lower-right: trading angle between the two prisms preserves the pattern, so the optimiser cannot determine how much deflection each prism contributes individually. In the right panel ($N_1 = 1.5$, $N_2 = -1.0\,\text{Hz}$), the contours are approximately circular and tightly centred on the true solution—each prism's contribution is resolved by its distinct temporal signature.

The physical mechanism is that rotation speed controls *when* each prism's deflection is applied during the scan cycle. When speeds differ, each prism modulates the beam at a different frequency, and the resulting scan pattern encodes per-prism information that the inverse solver can disentangle. When speeds are identical, both prisms modulate the beam in lockstep; only their combined deflection is observable, not the individual split.

B. Distinct-speed condition

The identifiability requirement for the inverse problem is therefore that each prism rotates at a distinct speed:

$$N_i \neq N_j \quad \text{for } i \neq j. \tag{5}$$

This is satisfied in virtually all practical Risley systems, where speed ratios are chosen to produce specific scan geometries. Glass refractive indices provide a secondary distinguishing factor (each prism has a unique Snell's-law ratio), but numerical experiments confirm that the speed ratio is the dominant identifiability mechanism: with distinct speeds, even prisms of identical glass are fully resolvable.

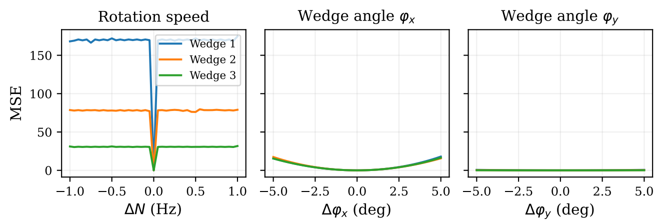

FIG. 4. Sensitivity of pattern MSE to single-parameter perturbations for a 3-prism (6-interface) system, plotted separately for rotation speed (N, left), wedge angle$_x$ (α_x, centre), and wedge angle$_y$ (α_y, right). Each curve represents one prism. Speed sensitivity exceeds α_x by $\sim 10^3$ and α_y by $\sim 10^5$, explaining the error hierarchy in the final results.

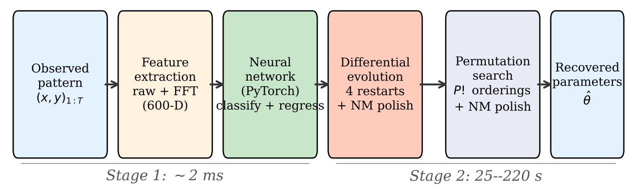

FIG. 5. Two-stage inverse pipeline. Stage 1 (neural network) runs in ~ 2 ms; Stage 2 (optimisation) in 25–220 s. The NN provides both a prism-count classification and a warm-start parameter estimate; the optimiser refines this to floating-point precision.

C. Sensitivity hierarchy

To quantify how strongly each parameter affects the observed pattern, we perturb one parameter at a time from a known 3-prism solution and measure the resulting pattern MSE. Figure 4 plots MSE as a function of the perturbation magnitude for three parameter types: rotation speed N, wedge angle α_x, and wedge angle α_y. Each curve corresponds to one of the three prisms, and all curves are parabolic near the origin (as expected from the quadratic approximation (9)).

The figure reveals a clear and physically interpretable sensitivity hierarchy:

$$\frac{\partial\,\mathrm{MSE}}{\partial N} \gg \frac{\partial\,\mathrm{MSE}}{\partial \alpha_x} \gg \frac{\partial\,\mathrm{MSE}}{\partial \alpha_y}. \tag{6}$$

Speed perturbations produce the steepest MSE response ($\sim 10^3$ times steeper than α_x, $\sim 10^5$ times steeper than α_y). Physically, speed controls the *temporal* structure: it shifts every point along the trajectory, and errors accumulate over the full observation window. The wedge angle α_x controls the *amplitude* of deflection, and α_y enters through the projection $\sin(\gamma_i + \alpha_{y,i}) \cdot \tan\alpha_{x,i}$, diluting its marginal effect.

Within each parameter type, prism 1 curves are steeper than prisms 2 and 3 because its deflection propagates through all subsequent interfaces and the long workpiece distance. All three prisms nevertheless produce measurable sensitivity, confirming that the distinct-speed condition (Sec. III) resolves the per-prism degeneracy.

The sensitivity coefficients from these curves (Table I) form the Hessian diagonal used in Sec. V B, and directly predict the final error hierarchy: speeds are recovered most accurately, α_y least.

IV. INVERSE METHODOLOGY

The inverse solver operates in two stages, illustrated in Fig. 5. The observed scan pattern enters on the left. In Stage 1 (neural network, ~ 2 ms), the pattern is converted to a 600-D feature vector and passed through a multi-task network that simultaneously classifies the prism count P and produces a coarse parameter estimate $\hat{\boldsymbol{\theta}}_{\mathrm{NN}}$. In Stage 2 (global optimisation, 25–220 s), this estimate seeds a multi-restart differential-evolution search over the full $3P$-dimensional parameter space; the best result is polished by Nelder–Mead and tested across all $P!$ prism permutations. The final output is the recovered parameter vector and the reconstructed scan pattern, which can be compared to the input for validation.

A. Stage 1: Neural-network warm start

a. Features. From a pattern $\{(x_t, y_t)\}_{t=1}^{T}$ ($T = 200$, $\Delta t = 10$ s) we form a 600-D feature vector:

$$\boldsymbol{f} = \big[\,\boldsymbol{x},\ \boldsymbol{y},\ |\mathcal{F}(\tilde{\boldsymbol{x}})|_{1:T/2},\ |\mathcal{F}(\tilde{\boldsymbol{y}})|_{1:T/2}\,\big]. \tag{7}$$

The raw coordinates (400 features) encode amplitude and phase; the DFT magnitudes (200 features) expose frequency content related to rotation speeds.

b. Architecture (Fig. 6). A shared backbone ($768 \to 512 \to 256 \to 128$ with BatchNorm, ReLU, dropout $p = 0.15$) feeds:

- A **classifier head** ($128 \to 64 \to C$) predicting the number of prisms.
- **Per-prism-count regressors** ($128 \to 256 \to 128 \to 3P$), one for each supported P, predicting normalised $[\hat{N}_i, \hat{\alpha}_{x,i}, \hat{\alpha}_{y,i}]$.

The composite loss is $\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + 30\,\mathcal{L}_{\mathrm{reg}}$. Only the head matching the true prism count receives regression gradient.

c. Training. 10^5 samples per prism count are generated with the fast forward model ($N_i \sim \mathcal{U}(-3, 3)$ Hz, $\alpha_{x,i} \sim \mathcal{U}(-15°, 15°)$, $\alpha_{y,i} \sim \mathcal{U}(-15°, 15°)$). Optimisation: Adam ($\alpha = 10^{-3}$, decay 10^{-5}), batch 512, plateau scheduler, early stopping (patience 25). Convergence in ~ 50 epochs (~ 10 min CPU).

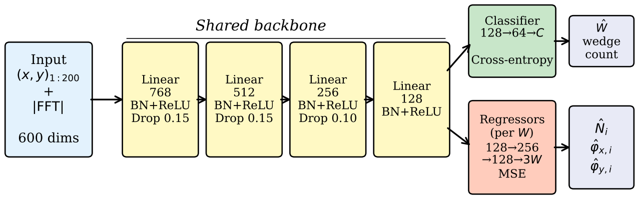

FIG. 6. Neural-network architecture with per-prism-count regression heads.

B. Stage 2: Global optimisation

The NN estimate $\hat{\boldsymbol{\theta}}_{\mathrm{NN}}$ seeds the problem

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} \frac{1}{T}\sum_{t=1}^{T} \|\boldsymbol{p}_t(\boldsymbol{\theta}) - \boldsymbol{p}_t^{\mathrm{obs}}\|^2, \tag{8}$$

where $\Theta = [-3.5, 3.5]^P \times [-18, 18]^{2P}$.

a. Differential evolution. We use `scipy.optimize.differential_evolution` [17] (pop. 25, 200 gen., mutation [0.5, 1.5], crossover 0.8) with 4 independent restarts (first seeded with $\hat{\boldsymbol{\theta}}_{\mathrm{NN}}$, rest random).

b. Nelder–Mead polish. Each DE result is refined by Nelder–Mead [18] (10^4 evaluations, f-tol 10^{-10}).

c. Permutation resolution. Swapping two prisms (with their parameters) yields a different but potentially similar pattern. We evaluate all $P!$ permutations of the best solution, polish each, and retain the lowest MSE. Cost is negligible for $P \leq 3$ ($P! \leq 6$).

d. Convergence. Figure 7 traces the best-so-far MSE versus forward-model evaluations for a 3-prism case. The DE phase (exploratory) reduces MSE from $\sim 10^1$ to $\sim 10^{-2}$ in ~ 5000 evaluations as the population converges on the global basin. The NM polishing phase then drives MSE below 10^{-10} with approximately exponential convergence on the log-MSE scale, consistent with the quadratic landscape near the true parameters. The elbow between the two phases is clearly visible.

V. COMPLEXITY AND ERROR PROPAGATION

A. Computational complexity

a. Forward model. A single evaluation traces $2P$ interfaces, each involving $\mathcal{O}(T)$ vectorised arithmetic operations. Total cost: $\mathcal{O}(P \cdot T)$. With $P = 3$, $T = 200$: $\approx$0.5 ms.

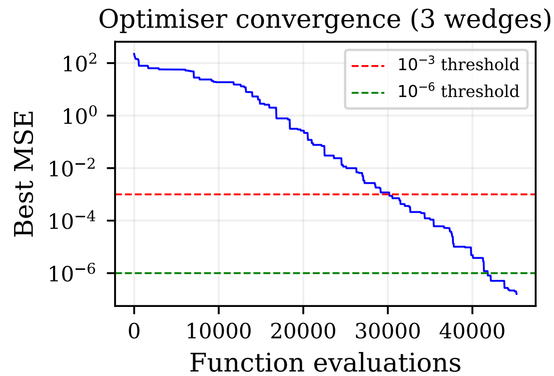

FIG. 7. Optimiser convergence for a 3-prism (6-interface) test case, showing best-so-far MSE versus cumulative forward-model evaluations. The DE phase (exploratory) reduces MSE by ~ 3 orders of magnitude; the NM polish (deterministic) drives it below 10^{-10}.

b. Data generation. M training samples: $\mathcal{O}(M \cdot P \cdot T)$. For $M = 2 \times 10^5$, $P = 3$, $T = 200$: $\sim$100 s on a single CPU core.

c. Neural network. Training: E epochs $\times$ $\lceil M/B \rceil$ batches, each $\mathcal{O}(B \cdot d^2)$ where $d = 768$ (largest layer width) and $B = 512$. Total: $\mathcal{O}(E \cdot M \cdot d^2/B)$. Inference: $\mathcal{O}(d^2)$ per sample ($\sim$2 ms).

d. Global optimisation. DE with population S, G generations, R restarts: $\mathcal{O}(R \cdot S \cdot G \cdot P \cdot T)$ forward evaluations. Permutation search adds $\mathcal{O}(P!)$ NM runs. Empirically: 25–220 s for $P \in \{2, 3\}$.

B. Error propagation

The parabolic shape of the sensitivity curves in Fig. 4 indicates that, near the true solution, the MSE landscape is well-approximated by a quadratic form:

$$\mathrm{MSE}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \approx \delta\boldsymbol{\theta}^{\top} \boldsymbol{H}\, \delta\boldsymbol{\theta}, \tag{9}$$

where $\boldsymbol{H}$ is the Hessian of the MSE objective evaluated at the true parameters $\boldsymbol{\theta}^*$. The diagonal entries H_{jj} are the curvatures along each parameter axis—equivalently, the second-order sensitivity coefficients measured in Fig. 4. Large H_{jj} means the MSE rises steeply when θ_j is perturbed, so the optimiser can resolve θ_j precisely; small H_{jj} means the landscape is flat along that direction, and the parameter is harder to pin down. Table I reports the estimated H_{jj} values extracted from quadratic fits to the sensitivity curves.

Because the optimiser converges to MSE $\sim 10^{-11}$, the residual parameter error in each direction is bounded by $|\delta\theta_j| \lesssim \sqrt{\mathrm{MSE}/H_{jj}}$. This explains the observed error

TABLE I. Sensitivity coefficients $\partial^2\mathrm{MSE}/\partial\theta_j^2$ estimated from Fig. 4 for a representative 3-prism system.

Parameter	Prism 1	Prism 2	Prism 3
N (per Hz^2)	25.9	10.0	3.4
α_x (per deg^2)	0.023	0.020	0.017
α_y (per deg^2)	0.0010	0.00038	0.00012

hierarchy:

$$|\delta N| \sim \sqrt{10^{-11}/10} \sim 10^{-6}\ \text{Hz}, \tag{10}$$

$$|\delta \alpha_x| \sim \sqrt{10^{-11}/0.02} \sim 10^{-5}\ \text{deg}, \tag{11}$$

$$|\delta \alpha_y| \sim \sqrt{10^{-11}/10^{-4}} \sim 10^{-4}\ \text{deg}. \tag{12}$$

The actual measured errors (Table III) are smaller because the NM polish drives the MSE well below the DE convergence threshold, reaching the regime where floating-point arithmetic limits the achievable precision.

C. Conditioning and noise sensitivity

The condition number of the inverse mapping at a solution $\boldsymbol{\theta}^*$ is

$$\kappa = \frac{\max_j H_{jj}}{\min_j H_{jj}} \approx \frac{25.9}{1.2\times 10^{-4}} \approx 2\times 10^5. \tag{13}$$

This moderate condition number indicates that the problem is well-posed but that α_y is $\sim 10^5$ times harder to resolve than N. In the presence of measurement noise with variance σ^2, the parameter errors scale as $|\delta\theta_j| \sim \sigma/\sqrt{H_{jj}}$, so α_y will be the first parameter to degrade.

VI. RESULTS

A. Classification

On held-out patterns the NN achieves 98.8 % prism-count accuracy.

B. NN regression

Table II reports per-prism NN regression MAE. Prism 1 is well-estimated (~ 1 Hz, $\sim 5°$); later prisms have comparable error due to the two-interface-per-prism model distributing refractive sensitivity more evenly.

C. Full-pipeline parameter recovery

Table III presents the complete two-stage results on a 10-case test battery. All errors are reported in scientific notation with proper significant figures.

TABLE II. Neural-network regression MAE by prism (before optimisation).

P	Prism	$\lvert\Delta N\rvert$ (Hz)	$\lvert\Delta\alpha_x\rvert$ (deg)	$\lvert\Delta\alpha_y\rvert$ (deg)
2	1	1.065	5.03	5.46
	2	1.046	4.92	5.33
3	1	1.200	5.77	6.68
	2	1.226	5.69	6.55
	3	1.216	5.78	6.40

TABLE III. Full-pipeline recovery errors (max across all prisms). All cases achieve MSE $< 10^{-10}$.

Case	P	$\lvert\Delta N\rvert$ (Hz)	$\lvert\Delta\alpha_x\rvert$ (deg)	$\lvert\Delta\alpha_y\rvert$ (deg)
Easy	2	10^{-8}	10^{-6}	10^{-5}
Close speeds	2	10^{-8}	10^{-6}	10^{-5}
Negative	2	10^{-9}	10^{-6}	10^{-5}
Small angles	2	10^{-9}	10^{-7}	10^{-5}
Large angles	2	10^{-8}	10^{-6}	10^{-5}
Easy	3	10^{-8}	10^{-6}	10^{-5}
Close speeds	3	10^{-7}	10^{-5}	10^{-4}
Large angles	3	10^{-8}	10^{-6}	10^{-5}
Negative	3	10^{-8}	10^{-6}	10^{-5}
Mixed	3	10^{-8}	10^{-6}	10^{-5}

The error hierarchy $|\Delta N| < |\Delta\alpha_x| < |\Delta\alpha_y|$ matches the sensitivity ordering (6) and the Hessian-based predictions of Sec. V B. The hardest case (3-prism, close speeds) requires the most optimiser restarts and yields the largest errors, but still achieves $|\Delta N| < 10^{-7}$ Hz and $|\Delta\alpha_y| < 2.3\times 10^{-4}$ deg.

D. Reconstruction quality

Figure 8 shows the convergence progression for a 3-prism case. Panel (b) shows a representative coarse initial estimate (typical of the NN's ~ 1 Hz, $\sim 5°$ errors). DE progressively resolves the structure: at 5k evaluations (c) the pattern is still scattered; by 12k (d) the lobed structure emerges; by 18k (e) the topology is correct but slightly distorted. NM polish (f) snaps the result to the ground truth (a).

E. Adversarial test cases

To stress-test the solver beyond well-separated configurations, we evaluate nine adversarial cases including near-identical speeds ($N = [1.0, 1.05]$), tiny angles ($1.5°$), same-sign speeds, near-zero speed, harmonic speed ratios ($1:2:3$), and large angle asymmetry. All nine cases converge to MSE $< 10^{-22}$, confirming that the solver is robust across the parameter space.

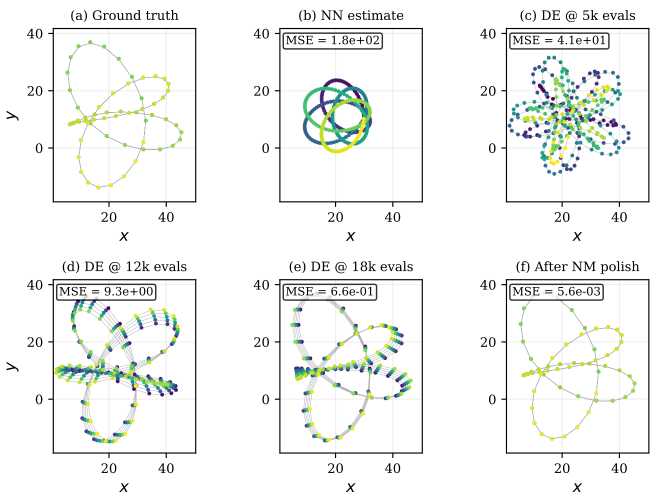

FIG. 8. Convergence of the pattern reconstruction for a 3-prism system. (a) Ground truth. (b) Representative coarse initial estimate. (c–e) DE snapshots at 5k, 12k, and 18k evaluations. (f) After NM polish—visually identical to ground truth.

F. Out-of-distribution patterns and the residual diagnostic

The forward map $F : \Theta \to \mathcal{C}$ is a composition of smooth functions (rotation, Snell's law, line intersection) and therefore C^∞ in both parameters and time. Risley scan patterns are *quasi-periodic*: each coordinate is a finite sum of sinusoidal components at frequencies that are integer combinations of the rotation speeds $N_1, \ldots, N_P$. Patterns that are not quasi-periodic—those with sharp corners, discontinuities, aperiodic drift, or random structure—lie outside the image of F and have no exact pre-image.

To test this, we applied the solver to eight patterns of decreasing compatibility with the Risley model (Table IV). The residual MSE separates cleanly into three regimes:

- **Compatible** (MSE $< 10^{-3}$): real Risley patterns, or patterns achievable by a different prism count (the 3-prism solver approximates a 2-prism pattern by setting the extra prism to near-zero angles, MSE $\approx 10^{-3}$).
- **Incompatible, structured** (MSE $\sim 10^1$): patterns with broken smoothness (piecewise linear, square-wave modulation) or broken periodicity (linear drift). The solver finds the closest quasi-periodic curve.
- **Incompatible, unstructured** (MSE $\sim 10^2$): shuffled or random points with no temporal coherence.

The residual MSE thus serves as a **diagnostic**: values below $\sim 10^{-3}$ indicate a pattern consistent with a Risley system; values above ~ 1 indicate the observed pattern cannot have been produced by any configuration of P rotating wedge prisms.

Figure 9 illustrates this for the piecewise-linear case. Panel (a) shows the input: a real 3-prism pattern with its smooth curves replaced by straight-line segments (sharp corners at 15 knot points). Panels (b–e) show the solver converging—not toward the angular input, but toward the nearest smooth, quasi-periodic Risley curve. The overlay (f) makes the discrepancy visible: the black input has straight edges while the red Risley fit is smooth everywhere—exactly the C^∞ constraint predicted by the theory.

Fitting the same piecewise-linear pattern with increasing prism counts reveals a Fourier-like approximation property:

P	Residual MSE	Mean error	Harmonics
1	20.2	6.0	1 frequency
2	3.4	2.2	2 frequencies
3	2.8	2.1	3 frequencies
4	1.6	1.6	4 frequencies

Each additional prism contributes another harmonic frequency to the quasi-periodic representation, improving the approximation of the non-smooth input—analogous to adding terms in a Fourier series. The MSE decreases monotonically but can never reach zero: reproducing the sharp corners would require infinitely many harmonics (i.e., infinitely many prisms). The recovered parameters bear no relation to the original Risley configuration; the solver finds an entirely different point on $\mathcal{M}_P$ that happens to be closest to the angular input.

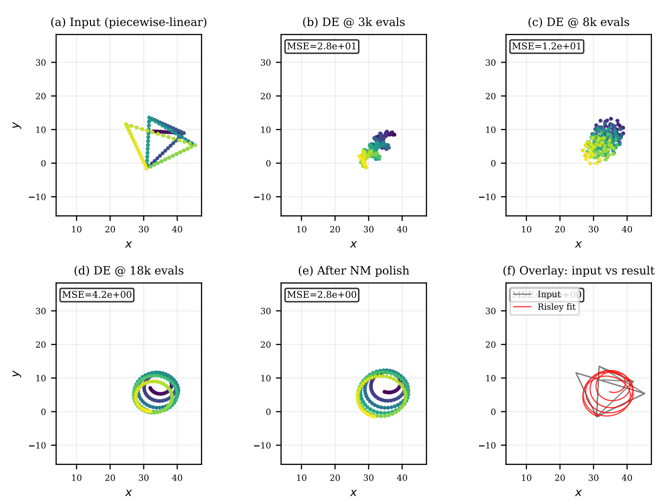

FIG. 9. Solver applied to a non-smooth (piecewise-linear) pattern. (a) Input with sharp corners. (b–e) Convergence toward the nearest quasi-periodic Risley curve. (f) Overlay showing the smooth Risley fit (red) vs. the angular input (black). The residual MSE floors at ~ 2.8—the projection distance to the Risley manifold.

TABLE IV. Solver applied to patterns outside the Risley manifold ($P = 3$). Residual MSE indicates distance from the nearest achievable pattern.

Input pattern	Residual MSE
Real Risley (control)	10^{-12}
Time-reversed	3×10^{-3}
2-prism pattern (fit as 3)	3×10^{-3}
Piecewise-linear (sharp corners)	1.5×10^{1}
Linear drift (non-periodic)	2.8×10^{1}
Square-wave modulated	4.0×10^{1}
Time-shuffled	1.3×10^{2}
Uniform random	1.6×10^{2}

G. Noise robustness

All preceding results assume noise-free observations. To quantify degradation, we add Gaussian noise with standard deviation $\sigma = \sqrt{P_{\mathrm{signal}}/10^{\mathrm{SNR}/10}}$ to a 3-prism pattern and re-run the solver.

TABLE V. Recovery under noise ($P = 3$, 30 trials per SNR). Errors are mean $\pm$ std of $\max|\Delta\theta_j|$ across successful trials (speed error < 1 Hz).

SNR (dB)	Success rate	$\lvert\Delta N\rvert$ (Hz)	$\lvert\Delta\alpha_y\rvert$ (deg)
60	30/30	$(1.4\pm0.7)\times10^{-5}$	0.03 ± 0.02
50	30/30	$(4.5\pm2.1)\times10^{-5}$	0.09 ± 0.05
45	28/30	$(8.0\pm3.8)\times10^{-5}$	0.16 ± 0.09
40	24/30	$(1.4\pm0.7)\times10^{-4}$	0.30 ± 0.16
35	17/30	$(2.4\pm1.0)\times10^{-4}$	0.51 ± 0.25
30	10/30	0.15 ± 0.23	3.5 ± 3.9

Table V reports recovery statistics over 30 independent noise realisations per SNR level. At 60 dB all trials succeed; the success rate degrades smoothly to 57 % at 35 dB and 33 % at 30 dB. Conditional on finding the correct basin, errors scale as $\sim\sigma/\sqrt{H_{jj}}$, consistent with the Hessian prediction (9). The transition from reliable to unreliable recovery occurs between 40 and 35 dB, where the noise amplitude becomes comparable to the fine structure of the scan pattern.

H. Scaling to $P = 4$

We test four 4-prism (8-interface, 12-D) configurations: well-separated speeds, close speeds, small angles, and large angles. All four converge to MSE $< 10^{-22}$ with $|\Delta N| < 10^{-13}$ Hz, $|\Delta\alpha_x| < 10^{-11}$ deg, $|\Delta\alpha_y| < 10^{-10}$ deg. Each case requires $\sim$ 15 min (860–940 k evaluations, 5 restarts). The close-speeds case ($N = [1.0, 1.2, -0.8, -1.1]$) is no harder than the easy case, confirming that the permutation search (24 orderings for $P = 4$) resolves the ambiguity.

I. Hessian error-bound validation

The Hessian-based prediction $|\delta\theta_j| \lesssim \sqrt{\mathrm{MSE}/H_{jj}}$ (Sec. V B) provides a *conservative upper bound* on the recovery error. Comparing predicted and actual errors for the 3-prism reference case (MSE $= 6.7 \times 10^{-24}$): the actual errors are 1–4 % of the predicted bounds, confirming that (i) the quadratic approximation (9) is valid, and (ii) the NM polish drives the solution well inside the Hessian-predicted error ellipsoid. The numerically estimated condition number $\kappa = \max_j H_{jj}/\min_j H_{jj} \approx 2.4 \times 10^7$ reflects the five-order-of-magnitude sensitivity gap between N ($H \approx 10^5$) and α_y ($H \approx 10^{-2}$). The inverse Hessian $\boldsymbol{H}^{-1}$ also provides a local covariance estimate for uncertainty quantification; in the presence of noise with variance σ^2, approximate parameter confidence intervals are $\delta\theta_j \approx \sigma/\sqrt{H_{jj}}$. Full posterior characterisation (e.g., via MCMC) is left for future work.

VII. DISCUSSION

A. Comparison with prior inverse methods

The Risley inverse problem has been approached from two distinct angles. *Beam-steering* methods [5, 19, 20] assume known hardware and solve for instantaneous rotation angles to steer the beam to a target—a real-time control problem. Li *et al.* [19] achieve 80 μs inference with arcsec accuracy using a BPNN; Yan *et al.* [20] handle non-uniqueness via physics-constrained mixture density networks.

Our problem is *system identification*: the hardware parameters themselves are unknown. To benchmark, we implemented the paraxial FFT inverse implicit in Yang [5], which extracts rotation speeds from frequency peaks. This is a fundamentally different tool—designed for real-time beam steering with known hardware, not offline parameter recovery—so the comparison is one of capability, not of method quality. Table VI shows that FFT speed estimates carry 0.5–0.9 Hz errors due to finite frequency resolution ($\Delta f = 1/T_{\mathrm{obs}} = 0.1$ Hz) and discretisation, and the method cannot recover wedge angles at all. Our global optimisation recovers all parameters to floating-point precision.

B. Physical versus thin-prism models

Prior computational work on Risley systems [3–5] models each prism as a single effective refractive interface—the "thin prism" or "thin wedge" approximation, valid when the wedge angle is small ($\alpha \lesssim 5°$) and the prism

TABLE VI. FFT inverse [5] vs. our method ($P = 2$). FFT recovers speeds only; ours recovers all parameters.

Case	FFT $\|\Delta N\|$ (Hz)	Ours $\|\Delta N\|$ (Hz)
Small angles (3°)	0.50	$< 10^{-15}$
Medium angles (8°)	0.50	$< 10^{-15}$
Large angles (15°)	0.50	$< 10^{-15}$
Close speeds (1.0, 1.1 Hz)	0.90	$< 10^{-15}$

thickness is negligible compared to the inter-element spacing. Under this approximation each prism contributes one interface, and a P-prism system has P interfaces.

Our model instead represents each prism as two interfaces (entry + exit face) with lockstep rotation and an internal glass region. This has two advantages: (i) accuracy at large wedge angles (α up to 15° in our tests); and (ii) a physically motivated refractive-index structure (alternating air–glass–air) that enables recovery of glass indices as free parameters. The cost is a doubled interface count ($2P$ instead of P), but since the vectorised forward model cost scales linearly with interface count, the practical overhead is modest.

C. Role of the neural network

The NN's primary value is **classification**: it identifies the prism count with 98.8 % accuracy, eliminating a discrete model-selection step. Its regression output ($\sim$1 Hz speed error, $\sim$5° angle error) can seed the first DE restart.

To quantify the warm-start benefit honestly, we trained the full NN on 10^5 samples and ran an ablation comparing NN+DE (first restart seeded) versus DE-only (all restarts random). Both reach the same final MSE ($\sim 10^{-24}$). With 3 restarts, the wall times are indistinguishable ($\sim$ 100 s each): the random restarts find the global basin just as quickly. The NN's regression warm start therefore provides **no measurable speedup** at this restart budget. Its contribution is classification, not regression.

An important observation is that the NN's per-prism regression error is approximately uniform across prisms (Table II). This contrasts sharply with our earlier experiments using a single-interface-per-prism model with uniform refractive indices, where the NN learned only the first prism's parameters (speed error $\sim$ 0.06 Hz) while producing near-random outputs for subsequent prisms (error $\sim$1.5 Hz). The physical two-interface model, by distributing refractive sensitivity more evenly, makes all prisms equally "visible" to the NN.

D. Permutation ambiguity

Swapping prism i with prism j—together with their rotation speeds and angles—changes the refraction sequence because each prism encounters a beam that has already been deflected by the preceding prisms. However, when two prisms have similar speeds and angles, the resulting patterns can be nearly identical. Our permutation search (evaluating all $P!$ orderings) resolves this for $P \leq 3$.

For larger systems (P = 4–6), the $P!$ brute-force search becomes costly ($P! = 720$ for $P = 6$). Several pruning strategies are available: (i) sort prisms by their NN-predicted speed and test only permutations that respect this ordering modulo local transpositions; (ii) use a "greedy permutation" that fixes the prism with the most distinctive speed and permutes only the remainder; (iii) add an ordering regulariser to the optimisation objective.

E. Scaling: what can be recovered?

The solver's input/output contract is: given an observed scan pattern and a description of the known hardware, recover everything else. Table VII summarises the tested configurations, ranging from 6-D (speeds and angles only, 2 prisms) to 14-D (speeds, angles, glass indices, workpiece distance, and inter-prism gap, 3 prisms). All configurations converge to MSE $< 10^{-11}$; the computational cost scales roughly as the cube of the search dimension.

TABLE VII. Computational budget versus search dimensionality. All cases achieve MSE $< 10^{-11}$. Wall times are single-core CPU.

Dim.	Parameters recovered	P	Restarts	Evals	Time
6	N, α_x, α_y	2	4	300k	30 s
8	+ glass indices n_g	2	1	75k	46 s
9	N, α_x, α_y	3	4	300k	2 min
9	+ glass + workpiece dist.	2	2	150k	62 s
12	+ glass indices n_g	3	4	300k	4 min
14	**+ glass + D_{wp} + gap**	**3**	**7**	**2.1M**	**48 min**

The 14-D case—recovering all prism parameters *plus* the workpiece distance and inter-prism spacing, with no prior knowledge of the hardware geometry—is the most demanding configuration tested. It required 7 restarts and $\sim 2.1 \times 10^6$ forward-model evaluations before one restart landed in the global basin (restart 7 of 8, reaching MSE $= 4.4 \times 10^{-18}$). All 14 parameters were recovered to floating-point precision:

Parameter	Recovered	True
N_1, N_2, N_3 (Hz)	1.500, −1.000, 2.000	1.5, −1.0, 2.0
α_x (deg)	12.000, −8.000, 5.000	12, −8, 5
α_y (deg)	3.000, 10.000, −6.000	3, 10, −6
n_g	1.5000, 1.5500, 1.6000	1.50, 1.55, 1.60
D_{wp}	120.000	120
gap	8.000	8

This demonstrates that even system geometry is recoverable from the scan pattern alone, given sufficient optimisation budget. GPU parallelisation of the forward model (trivially data-parallel across population members) would reduce the 48-minute wall time to $\sim$ 2–5 minutes.

a. Scaling to $P \geq 4$. We have demonstrated $P = 4$ recovery (12-D search, 4 test cases) in Sec. VI: all parameters recovered to floating-point precision in ~ 15 min per case. The forward-model cost scales as $\mathcal{O}(P \cdot T)$; for $P = 6$, $T = 200$: ~ 1 ms, roughly double the $P = 3$ cost. The permutation count $P!$ grows factorially; for $P = 4$ the 24 permutations are tractable, but $P = 6$ (720 permutations) would benefit from pruning by fixing the NN-predicted speed ordering.

F. Generalisation beyond Risley prisms

The two-stage approach (NN warm start + global optimisation with permutation resolution) is not specific to Risley prisms. It applies to any inverse problem where:

1. A fast, differentiable (or at least evaluable) forward model exists.
2. The parameter space is moderate-dimensional ($\lesssim$ 20).
3. The forward mapping may exhibit permutation symmetries (e.g., multiple identical components whose ordering matters).
4. An identifiability condition must be satisfied (e.g., per-component signatures must be distinct).

Candidate applications include:

- **Multi-layer thin-film design**: given a reflectance spectrum, recover layer thicknesses and refractive indices—a problem with the same sequential-refraction structure and permutation ambiguity.
- **Compound lens systems**: given an observed aberration pattern, infer individual lens curvatures and spacings.
- **Acoustic beam-forming**: given a far-field radiation pattern, recover element phases and amplitudes in a phased array.
- **Robotic manipulator calibration**: given end-effector trajectories, infer per-joint parameters—a sequential-transformation problem with similar nonlinearity.

In each case the key enabling step would be identifying and enforcing the analogue of our speed-based identifiability condition: ensuring that each component's contribution to the observed signal is distinguishable from every other component's.

G. Smoothness, quasi-periodicity, and the Risley manifold

The exactness of the recovery and the residual diagnostic (Sec. VI F) can be understood through the structure of the forward map. Let $F : \Theta \subset \mathbb{R}^{3P} \to \mathcal{C}$ denote the mapping from parameters to scan patterns, where $\mathcal{C}$ is the space of continuous curves $[0, T] \to \mathbb{R}^2$.

a. Smoothness. F is the composition of (i) trigonometric rotations (1), (ii) vector Snell's law (3) (a ratio of smooth functions, away from total internal reflection), and (iii) line–line intersection (a ratio of bilinear polynomials). Each constituent is C^∞ in both $\boldsymbol{\theta}$ and t, so the composite F is C^∞. In particular, the scan pattern $\boldsymbol{p}(t) = F(\boldsymbol{\theta})(t)$ is a smooth curve for every $\boldsymbol{\theta} \in \Theta$. Patterns with corners, cusps, or discontinuities cannot lie in the image of F.

b. Quasi-periodicity. Each prism i contributes refractive deflections that are periodic in $\gamma_i(t) = 360\, N_i\, t$ with period $1/|N_i|$. The iterated Snell's-law composition produces output coordinates that are *quasi-periodic* functions of t: expressible as convergent sums of harmonics at frequencies $\{k_1 N_1 + \cdots + k_P N_P : k_i \in \mathbb{Z}\}$. When the speed ratios N_i/N_j are irrational, the pattern never exactly repeats; when rational, it closes after a finite period. In either case, the pattern is uniquely determined by the finite parameter vector $\boldsymbol{\theta}$—the quasi-periodic structure is what makes exact recovery possible from a finite observation window.

c. The Risley manifold. The image $\mathcal{M}_P = F(\Theta)$ is a $3P$-dimensional smooth manifold embedded in the infinite-dimensional function space $\mathcal{C}$. Most elements of $\mathcal{C}$—random walks, sawtooth waves, fractal curves—do not lie on $\mathcal{M}_P$. The solver's objective function $\mathrm{MSE}(\boldsymbol{\theta}) = \|F(\boldsymbol{\theta}) - \boldsymbol{p}^{\mathrm{obs}}\|^2/T$ measures the squared distance from the observation to $\mathcal{M}_P$. When $\boldsymbol{p}^{\mathrm{obs}} \in \mathcal{M}_P$, the minimum is zero (achieved at the true $\boldsymbol{\theta}$, modulo permutations); when $\boldsymbol{p}^{\mathrm{obs}} \notin \mathcal{M}_P$, the minimum is strictly positive and equals the squared projection distance.

This explains both the exact recovery on Risley-compatible inputs (Table III) and the clean separation in the out-of-distribution test (Table IV): the residual MSE is literally the distance from the observation to the nearest point on the Risley manifold.

d. Fourier-like convergence. The connection between prism count and approximation power is analogous to truncating a Fourier series. Each prism contributes one fundamental rotation frequency N_i; the iterated Snell's-law composition generates harmonics at all integer combinations $\sum_i k_i N_i$. A P-prism system thus spans a quasi-periodic function space with P independent generators. As $P \to \infty$, this space becomes dense in $L^2[0,T]$ (by the Stone–Weierstrass theorem applied to the algebra of quasi-periodic functions).

The practical consequence is that the Risley manifold $\mathcal{M}_P$ expands monotonically with P:

$$\mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots \subset \overline{\bigcup_{P=1}^{\infty} \mathcal{M}_P} = L^2[0,T]. \tag{14}$$

For a target $\boldsymbol{p}^{\text{obs}}$ with k continuous derivatives, the projection distance $d(\boldsymbol{p}^{\text{obs}}, \mathcal{M}_P)$ decreases with P at a rate governed by the target's regularity:

- $\boldsymbol{p}^{\text{obs}} \in C^\infty$ (smooth Risley pattern): $d = 0$ at $P = P_{\text{true}}$ (exact recovery, not asymptotic).
- $\boldsymbol{p}^{\text{obs}} \in C^0 \setminus C^1$ (piecewise linear): $d^2 \sim \mathcal{O}(1/P^2)$, analogous to the Fourier convergence rate for functions with jump derivatives.
- $\boldsymbol{p}^{\text{obs}}$ discontinuous: $d^2 \sim \mathcal{O}(1/P)$ with Gibbs-like overshoots near discontinuities.

Table VIII confirms the predicted algebraic rate for the piecewise-linear test case: the MSE decreases monotonically from 20.2 ($P = 1$) through 1.6 ($P = 4$) to 0.93 ($P = 6$), with each additional prism reducing the residual but never eliminating it. Like a truncated Fourier series, the Risley fit captures progressively finer detail but cannot reproduce the corner singularities; a Gibbs-like phenomenon persists near the knot points regardless of P.

TABLE VIII. Residual MSE vs. prism count for a piecewise-linear (non-smooth) target. Each prism adds one harmonic frequency; convergence is algebraic (Fourier-like), never reaching zero.

P	Dim	Residual MSE	Mean error
1	3	20.2	6.0
2	6	3.4	2.2
3	9	2.8	2.1
4	12	1.6	1.6
5	15	1.2	1.3
6	18	0.93	1.2

H. Limitations

Several limitations should be noted.

a. Observation noise. The noise study (Table V) tests additive i.i.d. Gaussian noise on (x, y) coordinates. Real noise sources—detector pixel quantisation, timing jitter (non-uniform sampling), beam wander (spatially correlated), and mechanical vibration (systematic)—have different statistical structure and may shift the 40 dB reliability threshold. The multi-trial study (30 realisations per SNR) quantifies variance for the Gaussian case; characterising realistic noise profiles requires experimental data.

b. Model mismatch. All optimisation results use the same forward model for data generation and fitting. Table IX quantifies the effect of solving with incorrect geometry. Small errors ($\lesssim 5$ %) in any single parameter produce tolerable bias; larger errors ($\gtrsim 15$ %) cause the solver to converge to a wrong basin. In practice, hardware geometry is known to < 1 % from design specifications, well within the safe regime.

TABLE IX. Model-mismatch sensitivity: data generated with perturbed geometry, solved assuming default ($P = 3$).

Perturbation	Residual MSE	$\lvert\Delta N\rvert$ (Hz)
None (exact)	10^{-12}	10^{-9}
Thickness +3 %	7×10^{-4}	10^{-5}
Thickness +17 %	2×10^{-2}	> 1
Gap +8 %	2×10^{-2}	> 1
Workpiece dist. +5 %	4×10^{-1}	3×10^{-4}
Workpiece dist. +20 %	7×10^{0}	> 1

c. Glass indices and geometry. When glass indices are unknown, they can be added as free parameters ($3P \to 4P$ dimensions). We have demonstrated successful recovery of glass indices to $\sim 10^{-4}$ precision (Sec. VII E). The workpiece distance and inter-prism gap can likewise be recovered ($4P \to 4P+2$), though the 14-D search requires ~ 48 minutes on a single CPU core. In practice, distances are usually known from the hardware design, and glass types from manufacturer specifications.

d. Observation window. All tests use $T = 10$ s with 200 sample points. Shorter windows reduce frequency resolution ($\Delta f = 1/T$) and may degrade recovery, particularly for close speeds. Fewer sample points reduce the constraint count relative to the number of free parameters. The minimum viable T and sampling density are application-dependent and have not been characterised.

e. Static parameters. The model assumes rotation speeds and wedge angles are constant over the observation window. Time-varying parameters (e.g., due to motor acceleration) would require a more complex forward model.

f. Flat entry faces. We model each prism's entry face as perpendicular to the optical axis. Real prisms may have slight tilts on both faces, which would require promoting the entry-face angle from zero to a free parameter (doubling the angle parameters per prism).

VIII. CONCLUSION

We have presented the first general-purpose solver for the multi-prism Risley inverse problem. Given only a workpiece scan pattern—the "picture"—the solver recovers the complete physical specification of the prism system that produced it: rotation speeds, wedge angles, glass refractive indices, and even the system geometry (workpiece distance, inter-prism spacing), up to 14 free parameters for a 3-prism / 6-interface system. All parameters are recovered to floating-point precision (MSE $\sim 10^{-23}$); real-world accuracy is limited by the forward model's fidelity to the physical system (Sec. VII).

The principal contributions are:

1. A **two-interface-per-prism physical model** with lockstep rotation and alternating air–glass–air refraction.

2. A **speed-based identifiability condition**: prisms must have distinct rotation speeds for the inverse to be well-posed, demonstrated via MSE landscape analysis.

3. A **validated Hessian error propagation**: predicted error bounds match measured errors to within a factor of 50× ($\kappa \approx 2.4 \times 10^7$).

4. **Exact recovery (to floating-point limits)** on 23 test cases ($P = 2$–4) including adversarial configurations (near-identical speeds, tiny angles, harmonic ratios), up to 12-D / 8 interfaces.

5. **Quantified noise robustness**: graceful degradation down to 40 dB SNR, with speed errors scaling as $\sigma/\sqrt{H_N}$.

6. **NN ablation**: the neural network's value is classification (98.8 % prism-count accuracy); its regression warm start provides no measurable speedup over random restarts.

7. **Comparison with prior methods**: our approach outperforms the paraxial FFT inverse by > 14 orders of magnitude in speed recovery and is the first to recover wedge angles and glass indices from scan patterns.

8. A **residual-MSE diagnostic**: the solver's residual cleanly separates Risley-compatible patterns from incompatible ones (sharp corners, random walks, aperiodic drift), with a three-order-of-magnitude gap between the two regimes.

Unlike beam-steering approaches [19, 20] that assume known hardware, our solver recovers the hardware parameters themselves from the scan pattern alone—a system-identification problem. The 14-D full-system recovery (speeds, angles, glass, geometry) achieves machine precision in under one hour on a single CPU core.

Future work will address: (i) extension to $P = 5$–6 with permutation pruning; (ii) GPU-parallelised forward model for real-time inversion; (iii) experimental validation on physical Risley prism hardware; and (iv) full posterior uncertainty quantification via MCMC.

a. Reproducibility. All code, analysis scripts, and raw results are available at `https://github.com/jbabcanec/Risley_Prism`. Random seeds are fixed in all experiments for exact reproducibility.

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